

FITBUDDY AI

Your Personal AI Fitness Plan Generator

PROJECT DOCUMENTATION

A web-based fitness assistant that creates personalized workout, recovery, and meal guidance from user-provided fitness details.

Certificate

This is to certify that the project titled “FitBuddy AI – Your Personal AI Fitness Plan Generator” is prepared as an academic software project. The application collects basic user and fitness information and presents a generated fitness plan through a browser-based interface.

This documentation describes the project background, objectives, system design, modules, workflow, user interface, screenshots, testing approach, limitations, and future enhancements.

Declaration & Acknowledgement

The project is documented for academic and demonstration purposes. Supplied interface screenshots are the primary visual reference. Confirm exact Gemini model, API configuration, and validation behavior against final source code.

We thank our project guide, faculty, classmates, and supporters. Project theme: Personalized Fitness • Guided Workouts • Healthy Habits.

Abstract

FitBuddy AI is a web-based fitness plan generator. Its form collects name, age, weight, height, fitness goal, fitness level, and diet preference. Supplied generated content includes a seven-day workout schedule, exercise instructions, rest and recovery guidance, and balanced vegetarian meal suggestions.

The project concept uses generative AI with Google Gemini models and a Flask-based web application. Exact model name, endpoint, and deployment details must be verified from the final source code.

Keywords: Generative AI • Gemini • Fitness Plan • Workout Schedule • Flask • Personalized Guidance.

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1. Introduction

Fitness goals vary from person to person. A useful plan needs to consider a user's age, body measurements, goal, current fitness level, and food preference. Searching across separate sources for exercises, weekly schedules, recovery advice, and meal ideas can be confusing, especially for beginners.

FitBuddy AI brings these elements together in one interface. A user enters profile details and preferences, then receives structured guidance organized into workout, exercise instruction, recovery, and meal sections.

Target users include students, beginners, and individuals seeking a simple starting point for organizing general fitness habits. The system is an informational assistant and does not replace a qualified fitness or medical professional.

2. Background

Digital fitness tools have progressed from printed exercise charts and static workout applications to mobile trackers and AI-assisted recommendation interfaces. Generative AI can present information in natural language and organize it around user-provided context.

FitBuddy AI demonstrates this concept by collecting a small set of profile inputs and presenting a readable plan. The value of the application is the integration of input collection, generated content, and a single-page user experience.

3. Problem Statement

Users who are new to exercise may find it difficult to select suitable activities, arrange them across a week, understand exercise form, and coordinate activity with rest and food choices. Generic plans may not reflect the user's stated goal, experience, or diet preference.

Problem definition: develop a simple web application that accepts basic fitness details and returns a structured, understandable fitness plan while communicating that generated guidance is general and should be adapted safely.

4. Objectives and Scope

Objectives: collect user profile and preferences; generate a goal-oriented weekly workout outline; explain basic exercise movements; include recovery guidance; provide meal suggestions consistent with the selected diet preference; present the result in readable sections.

Scope: the demonstrated application focuses on plan generation and presentation. The supplied screenshots do not establish wearable integration, user accounts, long-term progress storage, medical screening, or live activity tracking.

5. Existing and Proposed System

Conventional approach: users combine advice from websites, videos, notes, or static workout charts. This can require manual selection and organization.

Proposed approach: enter profile details → submit the form → generate a structured plan → review schedule, exercise instructions, recovery notes, and meal ideas. The plan is intended as a starting guide rather than a guaranteed outcome.

6. System Requirements

Functional requirements: accept name, age, weight, height, fitness goal, fitness level, and diet preference; submit user information; display generated fitness guidance; organize output into sections; handle missing or invalid input with clear feedback.

Non-functional requirements: readable typography, responsive layout, understandable language, reasonable response time, secure handling of API credentials, and graceful handling of API/network errors.

Technology context: Python/Flask server and Google Gemini generative model integration are the project context supplied by the user. Confirm exact package versions, model ID, and front-end libraries from the final code before academic submission.

7. System Architecture and Workflow

Presentation layer: browser form and generated-plan view. Application layer: Flask route receives form data, validates it, constructs a prompt, and coordinates the request. AI service layer: Gemini model produces text based on the prompt. Response layer: the application renders the generated sections for the user.

Workflow: open FitBuddy AI; enter profile and preferences; submit; validate inputs; prepare the generation request; call the configured AI service; handle success or error; display the plan; user reviews the guidance.

8. Functional Modules

User Profile Module: captures name, age, weight, and height.

Goal and Preference Module: captures fitness goal, fitness level, and diet preference.

Plan Generation Module: sends relevant user inputs to the configured generative AI service.

Workout Schedule Module: presents a seven-day activity outline with active recovery and rest days.

Exercise Guidance Module: provides exercise names, sets/repetitions, and brief movement instructions.

Recovery Guidance Module: presents sleep, soreness, and pain-versus-effort reminders.

Meal Suggestion Module: displays meal ideas aligned with the chosen diet preference. The supplied example includes vegetarian meals.

9. Plan Generation Logic

The application uses the user's form values as context for generating recommendations. A well-formed prompt should request clear headings, a weekly schedule, beginner-appropriate exercise explanations, recovery notes, and food suggestions matching the selected diet.

Output quality depends on the prompt, model response, API availability, and validation performed by the application. The screenshots show generated prose but do not establish the exact prompt template or whether responses are post-processed.

Safety note: generated plans should avoid treating pain as a sign of progress. Users should stop an activity that causes sharp pain and seek qualified advice for injury, illness, or medical concerns.

10. User Interface Design

The supplied form uses a dark navy background with a centered card, green product title, clear labels, and wide input controls. The visible fields include Name, Age, Weight (kg), Height (cm), Fitness Goal, Fitness Level, and Diet Preference.

The generated output uses a dark card and readable text sections. In the screenshots, Markdown markers such as heading hashes and asterisks appear visibly in places; rendering Markdown into formatted headings and lists would improve presentation.

11. Application Screenshots

Figures below reproduce the supplied FitBuddy AI screenshots. They are included as visual evidence of the form and generated output. The screenshots are not altered to invent missing screens or features.



Figure 1. FitBuddy AI user input form

**FitBuddy AI**
Your Personal AI Fitness Assistant

Name

Hari haran

Age

19

Weight (kg)

64

Height (cm)

165

Fitness Goal

Weight Loss

Fitness Level

Beginner

Diet Preference

Vegetarian

Figure 2. Generated workout schedule and exercise instructions

2. Seven-Day Workout Schedule

This plan alternates between activity and rest to ensure your body has time to adapt.

- * **Day 1:** Full Body Strength (Bodyweight)
- * **Day 2:** Light Cardio (Brisk Walk) & Core
- * **Day 3:** Active Recovery (Light Stretching or Yoga)
- * **Day 4:** Full Body Strength (Bodyweight)
- * **Day 5:** Cardio (Jogging, Cycling, or Dancing)
- * **Day 6:** Lower Body & Core focus
- * **Day 7:** Full Rest Day

3. Exercise Names and Instructions

Perform each exercise for 2 sets of 10–12 repetitions.

- * **Bodyweight Squats:** Stand with feet shoulder-width apart. Sit back like you are sitting in an invisible chair. Keep your chest up and heels on the ground.
- * **Knee Push-ups:** Start in a plank position but drop your knees to the floor. Lower your chest toward the ground, then push back up. This builds upper body strength safely.

Figure 3. Rest, recovery, and vegetarian meal guidance

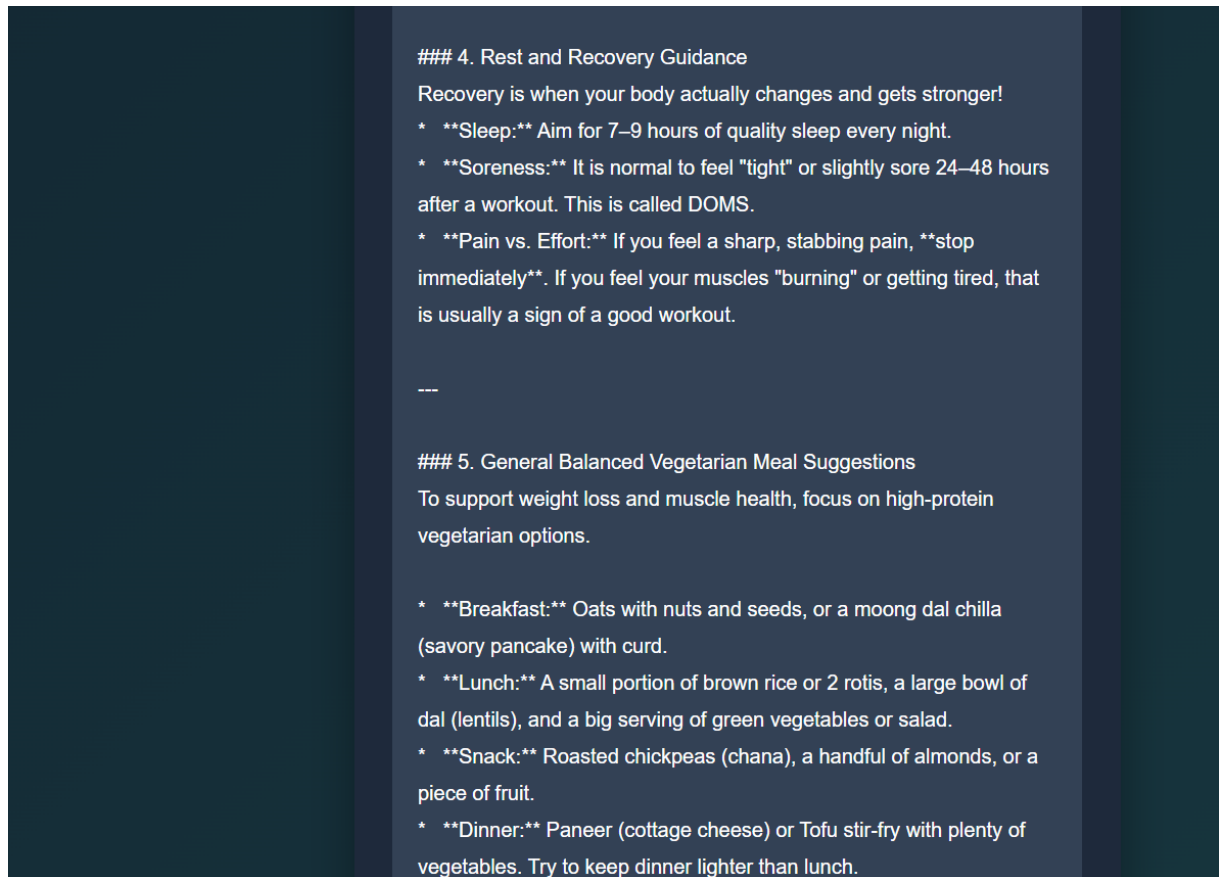


Figure 4. Additional generated plan content / interface view

12. Generated Plan Features

Seven-day schedule example: full-body strength, light cardio and core, active recovery, full-body strength, cardio, lower-body and core focus, and a full rest day.

Exercise example: bodyweight squats and knee push-ups are explained with basic setup and movement cues. The visible text specifies two sets of 10–12 repetitions.

Recovery example: the output discusses 7–9 hours of sleep, possible delayed-onset muscle soreness, and distinguishing ordinary exertion from sharp pain.

Vegetarian meal example: oats with nuts/seeds or moong dal chilla for breakfast; rice or roti with dal and vegetables for lunch; roasted chickpeas, almonds, or fruit for a snack; paneer or tofu with vegetables for dinner.

13. Testing and Validation

Recommended test cases: valid profile inputs produce a result; empty required fields show validation; non-numeric or out-of-range age/weight/height is handled; unsupported diet or goal values are rejected or safely normalized; API key missing produces a clear configuration error; API timeout or quota error produces a user-friendly message; generated content is displayed without broken formatting.

Testing status: the screenshots demonstrate the visible form and generated content. They do not independently prove all edge cases or backend/API reliability. Record actual pass/fail results after running the final application.

14. Results and Discussion

The supplied images demonstrate a FitBuddy AI input form and generated fitness-plan content, including a weekly workout outline, exercise instructions, recovery guidance, and vegetarian meal ideas.

The interface communicates the plan in one place. A remaining presentation improvement is to render Markdown syntax as formatted HTML and to provide clear loading, success, and error states during generation.

15. Limitations and Future Enhancements

Limitations: generated advice may be generic or inaccurate; no verified medical assessment is shown; no persistent progress history is evidenced in the screenshots; exact API/model configuration and error handling require source-code verification.

Future enhancements: Markdown rendering; input validation and safe range checks; user feedback and regeneration; downloadable PDF plan; progress tracking; optional activity integration; multilingual output; transparent AI disclaimer; secure environment-variable configuration; automated tests for API failures and edge cases.

16. Conclusion and References

FitBuddy AI demonstrates a generative-AI approach to organizing personalized fitness guidance. By combining profile inputs with a weekly workout outline, exercise instructions, recovery guidance, and meal suggestions, it provides a clear starting point for users planning general fitness routines.

References / project notes: supplied FitBuddy AI interface screenshots; final project source code and README; official Flask documentation; official Google Gemini API documentation for the exact SDK and model used; institutional report guidelines.